

MIHIR HADAVANI

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Software Engineer with hands-on experience building AI agent and Retrieval-Augmented Generation (RAG) systems – vector embeddings, semantic search, and agentic workflows – integrated with enterprise data via SQL and REST APIs. Comfortable across the decision-to-action lifecycle: data ingestion/ETL, validation and reconciliation, and production deployment with Python, Docker/Kubernetes, and cloud platforms.

Experience

SteepGraph

Jul 2025 – Present

Software Engineer

Pune District, Maharashtra, India (On-site)

- Architected a multi-agent AI system in **Agno** for Aras Innovator PLM that answers PLM data queries, safely authors schema and configuration changes under human-in-the-loop approval, and compares product structures/BOMs across versions – built on a Python/FastAPI + PostgreSQL/SQLModel backend.
- Built a custom **MCP (Model Context Protocol)** server with FastMCP exposing dozens of tools across Aras's OData REST API and raw AML, with per-session credential isolation, a self-invalidating schema cache, and a validation layer that reconciles Aras's case-sensitive list values before every write.
- Implemented a **RAG** pipeline on **pgvector** (hybrid search) with semantic chunking and embeddings, paired with an agentic memory layer – entity memory, a learned-knowledge store, and a decision log – so agents retain and reuse insights across sessions.
- Built a token-aware compression manager to summarize large tool results before they re-enter agent context, plus a disk-backed LLM response cache and an in-process, write-invalidated schema cache to cut redundant calls and latency.
- Integrated **Zitadel** for identity – a custom login UI over the Session API, PKCE authorization-code flow, RS256 JWT verification with rotating JWKS, and SSO/external-IdP login, plus machine-to-machine service auth via PAT and token introspection.
- Built an eval harness that replays real LLM calls against a live Aras instance to check tool-call ordering against intended playbooks, plus an automated test ensuring every new write tool ships behind the human-confirmation gate.
- Deployed the platform as a multi-tenant SaaS on **Docker** and **Kubernetes**, provisioning dedicated nodes per client for workload isolation, with **Traefik** as the reverse proxy/ingress and **Prometheus/Grafana** for per-tenant monitoring and alerting.
- Bring strong **software design principles** to a parallel **Java/Spring Boot** integration framework, applying Factory/Adapter patterns and Spring DI so new enterprise integration targets plug in with zero changes to the core dispatch logic.

Inventyv Software Services Pvt Ltd

May 2024 – Dec 2024

Devops Intern

Onsite

- Built a CI/CD pipeline with GitHub Actions, Docker, Spinnaker, and Kubernetes, automating build, test, and deployment for faster, more reliable releases.
- Tuned Kubernetes HPA policies and resource limits so the platform scaled cleanly under variable traffic.
- Set up Prometheus and Grafana dashboards for real-time monitoring, enabling proactive performance tuning and faster incident response.

Education

Dharmsinh Desai University, Nadiad

2021 - 2025

B.Tech In Information Technology

8.89 /10 CGPA

Modi School, Rajkot

2019 - 2021

HSC

91.53 Percentage

Projects

FinGuru: News Research Tool

[Source Code](#)

- Tech : LangChain, Python, Google Gemini, Streamlit, AWS Lambda, AWS Api Gateway, AWS SageMaker, AWS S3
- Built a Gen AI research tool that ingests URLs and PDFs, generates vector embeddings over the extracted text, and uses LangChain + Google Gemini to answer questions and surface insights grounded in the source articles.

Kavach: Fintech Resilience Control Plane

[Source Code](#)

- Tech : Java, Rust (Tonic, sqlx), gRPC, Apache Kafka, PostgreSQL, Docker Compose, System Design
- Designed Kavach to solve a real fintech reliability gap: a single BaaS-provider/bank pair failing or degrading halts money movement entirely, since apps today integrate directly with one provider stack with no resilience layer underneath.
- Authored a full HLD/LLD system design across 5 database-per-service microservices, applying saga-based failover orchestration, circuit breakers, CQRS-lite, idempotent-receiver, and event-carried state transfer; used gRPC for synchronous correctness-critical calls (breaker checks, ledger posting) and Kafka for async, at-least-once fact propagation (webhook normalization, ledger postings).

Technical Skills

Languages : Python, SQL, Java, TypeScript

AI / Gen AI : LLM APIs, RAG, Vector Embeddings (pgvector), Semantic Chunking, Prompt Engineering, Agentic Memory, Agno, MCP/FastMCP, LangChain

Data Engineering : PostgreSQL

Backend & System Design : FastAPI, Spring Boot, gRPC, Apache Kafka, Microservices, Design Patterns (Factory, Adapter), Dependency Injection

Identity, Cloud & DevOps : Zitadel (OIDC/PKCE/JWKS/SSO), AWS, Azure, Docker, Kubernetes, Traefik, Prometheus/Grafana, Git, GitHub Actions, Agile

Certificates/Achievements

<u>AWS Academy Graduate - AWS Academy Machine Learning Foundations</u>	May 2024
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<u>AWS Academy Graduate-AWS Academy Cloud Foundations</u>	December 2023
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<u>Google Cloud Study Jams</u>	November 2023
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<u>Full Stack Development</u>	November 2022
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Algorithmic Competitions

- 300+ Questions Solved on [LeetCode](#)